

ARITESH MAJI

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SUMMARY

Pre-final year Computer Science undergraduate (CGPA 8.30) with a backend-focused engineering profile spanning REST and GraphQL APIs, RAG systems, and high-performance data pipelines in Python. Strong foundations in data structures & algorithms and system design, with direct experience analyzing real-time financial market data through a multi-agent trading intelligence platform. Comfortable writing clean, modular Python code, working with APIs and data transformation pipelines, and debugging and optimizing scripts in fast-paced environments.

SKILLS

Languages: Python, C++, SQL

Data & Analytics: Data Structures & Algorithms, Real-Time Data Pipelines, Data Extraction & Transformation, Semantic Caching

Backend & APIs: FastAPI, REST APIs, GraphQL, Microservices, JWT Auth

Databases: MySQL, PostgreSQL, Oracle, Firebase, Redis, Milvus (Vector DB)

AI / LLM: RAG, LangGraph, Groq LLM, Hugging Face, LLM Automation, Prompting

Core CS: OOP, Operating Systems, System Design

Tools & Cloud: Docker, AWS, Git

INTERNSHIP EXPERIENCE

Information System Trainee | Indian Oil Corporation Limited 12/2025 – 01/2026 | Haldia, West Bengal, India

- Developed a scalable Python-based RAG chatbot, cutting manual file-search time by 80% for internal enterprise knowledge bases.
- Engineered a production-grade REST API voice assistant using Python and Whisper, reducing navigation latency by 40% across complex internal enterprise platforms.

PROJECTS

Chosa Agent — Multi-Agent Financial Intelligence Platform | Python, FastAPI, LangGraph, Groq LLM, Next.js 02/2026 – 03/2026

- Built a multi-agent platform in Python to analyze real-time market data, news, and portfolio activity, cutting manual analysis time by 80% and enabling data-driven investment decisions.
- Engineered high-concurrency FastAPI pipelines to ingest real-time market streams and integrated Groq LLMs for low-latency, portfolio-personalized recommendations with user-controlled decision constraints.

Semantic Rate-Limiting System for GraphQL APIs | Python, GraphQL, PostgreSQL, Redis, Docker 02/2026 – 03/2026

- Reduced API response latency by 50% by scripting semantic caching and query-cost optimization for repeated queries.
- Cut backend service load by 60–90% with complexity-aware throttling and middleware controls, reaching a 62–69% semantic cache hit rate.

Multi-Agent RAG Platform | Python, FastAPI, Milvus, BM25, SPLADE, OLLAMA, Docker 12/2025 – 01/2026

- Architected a "Fail-Closed" RAG system with Single Document Authority, eliminating hallucinations via sentence-level verifiable citations.
- Built a hybrid retrieval pipeline combining Milvus, BM25, and SPLADE with cross-encoder reranking, and secured data access with RBAC and metadata filtering.

Browser-Based Voice Assistant API | Python, Flask, Pydantic, Whisper 02/2025 – 04/2025

- Built a Flask API for voice-driven navigation, converting natural-language commands into actionable URLs using OpenAI Whisper.
- Hardened API security with JWT authentication, CORS, and Pydantic-based rate limiting; containerized with Docker for consistent deployments.

EDUCATION

Bachelor of Technology, Computer Science & Technology 2023 – Present

Institute of Engineering and Management, Kolkata, West Bengal, India | CGPA: 8.30 / 10

Coding Profile: 200+ problems solved on LeetCode

CERTIFICATIONS

- Gen AI and RAG Application — IBM
- Python for Data Science — IBM
- Information Theory — The Chinese University of Hong Kong
- Advanced System Security Topics — University of Colorado System

ACHIEVEMENTS

- Top-performing candidate at the ET Gen-AI Hackathon 2026.
- Received a Letter of Recommendation from the General Manager, Indian Oil Corporation, for high-quality code contribution and being a quick learner among all interns.